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$$\lim_{x\to\infty} \frac{\sqrt{x^2+3}}{2x^2-1}$$
$$= \lim_{x\to\infty} \frac{\sqrt{x^2\left(1+\frac{3}{x^2}\right)}}{x^2\left(2-\frac{1}{x^2}\right)}$$
$$= \lim_{x\to\infty} \frac{|x|\sqrt{1+\frac{3}{x^2}}}{x^2\left(2-\frac{1}{x^2}\right)}$$

$$\lim_{x\to+\infty} \frac{|x|\sqrt{1+\frac{3}{x^2}}}{x^2\left(2-\frac{1}{x^2}\right)} = \lim_{x\to+\infty} \frac{x\sqrt{1+\frac{3}{x^2}}}{x^2\left(2-\frac{1}{x^2}\right)} = 0$$
$$\lim_{x\to-\infty} \frac{|x|\sqrt{1+\frac{3}{x^2}}}{x^2\left(2-\frac{1}{x^2}\right)} = \lim_{x\to-\infty} \frac{-x\sqrt{1+\frac{3}{x^2}}}{x^2\left(2-\frac{1}{x^2}\right)} = 0$$

# References